

ZS-F29

Prime-Blind Lifts Cannot Generate Prime-Power Depth: A Koenigs–Abel Obstruction and a Local Height Model for Explicit-Formula Towers

with an application to the Z-Spin Riemann-Bridge programme (the Archimedean Vertical and the Six-Layer Ladder)

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Theme: Foundations / RH Bridge · Paper code: ZS-F29 · v1.2 (continuation of ZS-F24 → ZS-F25 → ZS-F26 → ZS-F27; terminal addendum to the closing release)

Verification: 30/30 PASS (zs_f29_verify_v1_2.py, mpmath 120 digits + exact rational arithmetic, pre-registered seed 20260612) | Zero Free Parameters | NO CLAIM on RH / GRH | Gate D5 remains IMPORTED-OPEN $\equiv$ RH | Corrections of record in §8.3 | Rate Spectrum No-Go and F30–F33 adjudication in §6.5–§6.7

§0. Abstract

A one-parameter unitary flow sampled at the prime reciprocals $\{1/p\}$ cannot be promoted to prime-power depth by any prime-independent auxiliary structure. We prove this in two layers. The Socle Obstruction (Theorem 4.6, PROVEN, stated for arbitrary groups): the subgroup of $\mathbb{Q}/\mathbb{Z}$ generated by $\{1/p : p \text{ prime}\}$ is exactly the socle $\oplus_p \mathbb{Z}/p$ — the elements of squarefree order — so for any group G and any homomorphism $\pi : G \rightarrow \mathbb{Q}/\mathbb{Z}$ whose generators all map into the socle, the image of π lies in the socle, and no element maps to $1/p^k$ for $k \geq 2$ (Squarefree Span Lemma 4.1; operator instance Theorem 4.3 on $\mathbb{C}^{11} \otimes K$). A spectral corollary, corrected in this release (Corollary 4.4' with a documented correction of record: the v1.0 proof step “the eigenvalue multiset determines t up to sign” is false — counterexample $\text{spec } U(1/11) = \text{spec } U(2/11)$, check G-21 — while the conclusion stands via the conjugation-invariant maximal order, check G-22), excludes conjugation routes; and the Universal-Multiplier Mismatch (Corollary 4.5) excludes rate-matching by the single Koenigs multiplier.

The positive layer has two parts. First, the Height Coordinate Theorem (Theorems 3.1/3.3, DERIVED from Koenigs 1884): since $|\lambda| = (\pi/2)|z^*| = 0.891514 < 1$, the i -tetration $T(z) = i^z$ carries a canonical Abel height α with $\alpha(i^z) = \alpha(z) + 1$ — the Z-Spin iteration is a unit vertical shift on an aperiodic archimedean axis with one universal rate $-\log|\lambda| = 0.114835$. Second, new in v1.1, the Local Height Model (Model 6.1, IMPORTED-known: the primon-gas / Bost–Connes single-prime factor): $H_p = (\log p) \cdot N$ on $\ell^2(\mathbb{N}_{\geq 1})$ realizes the explicit-formula support $\{k \log p : k \geq 1\}$ exactly, and the graded traces assemble to $\sum_p \sum_k (\log p) p^k \{-ks\} = -\zeta'(s)/\zeta(s)$ (checks G-24/G-25), executing the v1.0 pre-registered factorization audit with result PASS: the weight factorizes as $\text{rate}(p) \times \text{level}(k)$ (check G-26). The model thereby closes the v1.0 open problems O-F29.1/O-F29.2 at the model level and relocates their open content precisely: the level space is universal; the only per-prime datum is the rate $\log p$; and the corpus

owns exactly one rate — the archimedean one. The missing engine of the adelic locator is therefore not a space and not a shift but the family of rates $\{\log p\}$, whose corpus-internal derivation is the sharpened open problem O-F29.1'.

The Six-Layer Ladder of the Riemann engagement (§7) is upgraded from a consolidation table to a gated evaluation framework for locator models: frame and detection SHARED (DERIVED-interpretation / PROVEN); support CLOSED against the entire prime-blind class (Theorem 4.6) and realized at model level by import; weights CLOSED at model level, OPEN at derivation level; location/positivity OPEN with the internal scalar route PROVEN-closed (ADS-5/6); count axis a sharpened NON-CLAIM. Two corrections of record are carried (§8): the ZS-M23 §12 principle refined to its three audited route classes, and NC-M28.7 sharpened; §8.3 adds the Corollary 4.4 correction together with the protocol lesson — a proof-step adversarial audit is added to the verification standard, with Lemma 4.1/Theorem 4.6 registered as formalization candidates. Per the programme audit, this is the closing release of the F29 line: the paper is, by its own declared genre, a map, a correction, and an honest import — not a density contribution — and subsequent external effort is redirected to the seam-constrained Fuglede track. Verification: 30/30 PASS. Zero free parameters; A = 35/437 and Q = 11 LOCKED, A not consumed numerically. No claim on RH; Gate D5 remains IMPORTED-OPEN $\equiv$ RH.

Version 1.2, the terminal addendum, executes the decidable core of two further audits. The Rate Spectrum No-Go (Theorem 6.5, PROVEN) closes the declared-class branch of O-F29.1' in the negative by two one-line obstructions: $\text{Hom}(\text{socle}(\mathbb{Q}/\mathbb{Z}), (\mathbb{R}, +)) = 0$ — the corpus prime data is torsion, real rates are torsion-free, so every additive extraction vanishes — and $\dim_{\mathbb{Q}} \text{span}_{\mathbb{Q}} \{\log p\} = \aleph_0$ by unique factorization (numeric face: $\min |\sum a_p \log p| = 8.59 \times 10^{-5}$ over nonzero $|a_p| \leq 6$, check G-29), so no finite family of corpus constants $\mathbb{Q}$ -generates the rates. Relative to homomorphic and finite- $\mathbb{Q}$ -linear extraction, $\{\log p\}$ is necessarily imported as infinitely many independent data; the residue O-F29.1" (nonlinear, spectral, or analytic derivation) is, by Theorem 5.1, the locator construction itself. The semilocal height ansatz $H_S = H_{\infty} \oplus \oplus_{p \in S} (\log p)N$ is registered with its three gates — SG1 finite-S positivity, SG2 positivity-preserving convergence, SG3 test-class density — each mapped onto a previously registered wall (ZS-F26 rung 1; O-F26.5; classical): registration, not progress (NC-F29.10). Of four externally proposed RH-grade lines, the naive total-positivity route (F33) is REFUTED (Proposition 6.7a, PROVEN: the finite-place kernel has zero diagonal mass, two-point counterexample $-\sqrt{2} \log 2 = -0.9803 < 0$, check G-27; Bochner sign-indefiniteness of the smoothed kernel, grid minima -55.6 and -125.5 , check G-28; structural reductio: a positive-definite prime side would trivialize Weil positivity), while F30–F32 are declined as papers and absorbed as gates per the second audit's calibration. The programme portfolio is registered: $\approx 80\%$ seam-constrained Fuglede track, $\approx 20\%$ independent-verification interface with the truncated-Weil-form / prolate programme, with the $\sup \varepsilon^*(N) \leftrightarrow$ small-eigenvalue translation bridge as the single corpus $\leftrightarrow$ frontier interface, and anti-confabulation gate F-F29.12 binding on successors.

§0.1 Epistemic Status Legend

TAG	MEANING
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PROVEN	Rigorous proof under declared definitions; verified by direct computation where finite; no open step. From v1.1, every PROVEN tag additionally passes a proof-step adversarial audit (§8.3, §9.2).
DERIVED	Valid deduction from PROVEN results and imports; zero new free parameters.
DERIVED-interpretation	Structural synthesis of PROVEN/DERIVED components; the synthesis is new, the components are not.
IMPORTED-PROVEN	External peer-reviewed result used without re-proof; full citation.
IMPORTED-known	External mathematics that is standard or folklore in its community; cited, used without re-proof, and explicitly not claimed as new.
IMPORTED-OPEN	External open problem imported at its open status (Gate D5 $\equiv$ RH).
COMPUTED	Exact finite computation (exhaustive enumeration or pre-registered sampling).
HYPOTHESIS-strong	Motivated conjecture with explicit closure path; not derived.
OBSERVATION	Structural coincidence registered without mechanism; no evidentiary weight.
OPEN	Registered open problem.
NON-CLAIM	Explicitly not asserted.
RETIRED	Conjectured form falsified in-session and replaced by the surviving form.
CORRECTED	A published proof step found false under adversarial audit; the corrected proof and the defect are both recorded (no deletion).

§1. Introduction

Version 1.0 of this paper executed the Vertical Conversion reconnaissance registered against ZS-F27 O-F27.2 and reported a split verdict: the weak form of the hypothesis — a canonical vertical coordinate on which the Z-Spin iteration is a unit shift — survived as a theorem anchored to Koenigs (1884), while the strong form — that this verticality generates prime-power depth — died by the arithmetic of squarefree denominators, in its entire prime-blind class. Version 1.1 responds to two independent audits of v1.0 and is the closing release of the F29 line.

The first audit prescribed external positioning: lead with the statements an outside reader needs no corpus vocabulary to evaluate — why a single universal vertical shift cannot manufacture $\{1/p^2\}$ from $\{1/p\}$, and what structure does produce the towers $\{k \log p\}$ with von Mangoldt weights. v1.1 adopts this at zero claim-inflation cost: the obstruction is promoted from its operator instance to a group-extension statement in socle language (Theorem 4.6), and the constructive counterpart is supplied by an honest import — the primon-gas / Bost–Connes local factor (Model 6.1), which realizes support and weights simultaneously and thereby executes, with result PASS, the factorization audit this paper pre-registered in v1.0 §8. The second audit found a defect: the v1.0 proof of Corollary 4.4 contained a false step (eigenvalue-multiset rigidity up to sign), discovered by adversarial counterexample construction — $\text{spec } U(1/11) = \text{spec } U(2/11)$. The conclusion is unaffected; the corrected proof is simpler; and the defect, its counterexample, and the protocol lesson are recorded in §8.3 under the no-deletion convention. The same audit’s genre verdict is accepted in full: by the hardest-step test (§9.1), the most difficult ingredient of this paper is textbook (the socle of $\mathbb{Q}/\mathbb{Z}$; Koenigs 1884; the Euler product), so the paper is a map, a correction, and an import — not a density contribution — and it says so.

Genre and scope. §3–§6 are stated so that a reader with no knowledge of the Z-Spin corpus loses nothing: §3 is classical one-dimensional holomorphic dynamics; §4 is elementary arithmetic of $\mathbb{Q}/\mathbb{Z}$ and unitary one-parameter groups; §6 is standard arithmetic statistical mechanics, imported. The corpus interface begins at §5. Programme note (registered): per the focus audit, the F29 line completes with this release; no v2 is planned; subsequent external effort of the collaboration is directed to the seam-constrained Fuglede track (ZS-F28 line), whose own audit cycle proceeds in parallel. A paper-code disambiguation from v1.0 is preserved: the profinite-lift material of the seam programme exploration is not this paper and remains earmarked for a ZS-F28 revision appendix.

Terminology. Per the corpus terminology convention, the dynamical agent is the Z-Spin action: we write Z-Spin iteration, Z-Spin mediation, Z-Spin dynamics; the Z-sector is the stage on which the action operates.

§2. Locked Inputs and Imports

Table 2.1. Inputs and imports. A = 35/437 is LOCKED corpus-wide but is not consumed numerically in this paper; Q = 11 enters only as the register dimension through the imported flow. New v1.1 imports in the last three rows.

Object	Statement	Status
Fixed point z^*	$z^* = i^{\wedge}\{z^*\}$, unique attracting fixed point of $T(z) = i^{\wedge}z$ in the principal-branch basin; $z^* = 0.438283\dots + 0.360592\dots i$ (ZS-M1; checks G-01, G-04).	PROVEN (ZS-M1)
Multiplier λ	$\lambda = T'(z^*) = (i\pi/2) z^*$; $ \lambda = (\pi/2) z^* = 0.891514\dots < 1$ (checks G-02, G-03, G-05).	PROVEN (ZS-M1)
Register flow	$U(t) = e^{\wedge}\{2\pi i t D\}$, $D = \text{diag}(j - 5)$ on $\mathbb{C}^{11}$; $\text{Tr } U(t) = \sin(11\pi t)/\sin(\pi t)$ (ZS-F27 §4; checks G-10, G-11, G-19).	PROVEN (ZS-F27)
Transfer family	$W_{\text{p}} = U(1/p)$, $\text{order}(W_{\text{p}}) = p$; multiplicative closure $\oplus_{\text{p}} \mathbb{Z}/p$ with Pontryagin dual $\prod_{\text{p}} \mathbb{F}_{\text{p}}$ (ZS-F27 Prop. 4.1, Thm 4.2; check G-12).	PROVEN (ZS-F27)
Depth-One Obstruction	Register Wilson algebra realizes every prime at depth one; locator requires towers $\{k \log p\}$ with $\zeta_{\text{p}}(k)$ weights (ZS-F27 Thm 5.1; ZS-F25 Thms 6.4-N/6.4-W).	DERIVED (ZS-F27)
Finite Trace Rigidity	No finite-dimensional self-adjoint flow produces a Guinand–Weil logarithmic singular support (ZS-F27 Thm 5.5).	PROVEN (ZS-F27)
Archimedean detection	i -tetration pulsation is the archimedean scaling detector; adelic locator OPEN (ZS-F24 v2.0; ZS-F25 v2.0).	PROVEN / OPEN
Z_2 frame	$\sigma = 1/2 \leftrightarrow j = 1/2$ involution-fixed-locus isomorphism (ZS-M28 Thm 28.5).	DERIVED-interpretation
Internal scalar no-go	No V_4 -equivariant scalar achieves Weil positivity (ZS-M22 ADS-5/ADS-6).	PROVEN (ZS-M22)
Programme principle	“Z-Spin does not, and structurally cannot, prove RH from internal data alone” (ZS-M23 §12; refined §8.1).	Principle
Count-axis non-claim	No shared count-axis observables (ZS-M28 NC-M28.7, Cor. 28.5a; ZS-F13 Thm F13.2; sharpened §8.2).	NON-CLAIM / DERIVED-COND.
Koenigs linearization	Attracting fixed point with $0 < \lambda < 1$: $\exists$ unique normalized ϕ , $\phi \circ T = \lambda \phi$ on the basin [1, 5].	IMPORTED-PROVEN

Socle of $\mathbb{Q}/\mathbb{Z}$	The torsion elements of squarefree order form the subgroup $\oplus_p \mathbb{Z}/p$ (the socle); standard abelian-group theory [8].	IMPORTED-known
Primon gas / BC factor	$H = \sum_p (\log p) \cdot N_p$ has bosonic partition function $\zeta(s)$; single-prime graded trace $\log p \cdot p^{-s} / (1 - p^{-s})$; $\Sigma = -\zeta'/\zeta$ [9, 10, 11].	IMPORTED-known
Register dimension	$Q = 11$; $(Z, X, Y) = (2, 3, 6)$ (ZS-F5).	PROVEN; LOCKED

§3. The Height Coordinate of the Z-Spin Iteration (unchanged from v1.0; stated Z-Spin-free)

§3.1 Theorem 3.1 (Koenigs Linearization at z^*). [DERIVED from IMPORTED-PROVEN]

Let $T(z) = i^z$ on the principal branch and z^* its attracting fixed point. Since $\lambda = T'(z^*) = (i\pi/2)z^*$ satisfies $0 < |\lambda| = 0.891514\dots < 1$ (checks G-02, G-03), the Koenigs theorem [1, 5] applies: there exists a holomorphic ϕ on the immediate basin, unique under $\phi(z^*) = 0$, $\phi'(z^*) = 1$, with

$$\phi(T(z)) = \lambda \cdot \phi(z).$$

Numerical confirmation at $n = 900$ iterations, 120 digits: residual 3.6×10^{-47} at five basin points (checks G-06, G-07). [Verified]

§3.2 Definition 3.2 and Theorem 3.3 (Unit Vertical Shift). [DERIVED]

On the basin minus the zero set of ϕ define $\alpha(z) = \log \phi(z) / \log \lambda$. Then the Abel equation holds:

$$\alpha(i^z) = \alpha(z) + 1,$$

i.e. one Z-Spin iteration is a unit vertical ascent on a canonical, aperiodic height axis (checks G-08, G-20; increment residual 9.0×10^{-47}). The axis carries one universal rate, $-\log|\lambda| = 0.114835\dots$ Remarks 3.4–3.5 of v1.0 (founding-note coordinates; the two faces of “+1” — the finite horizontal shift S of $J = F^2S$ versus the infinite vertical Abel shift) are preserved verbatim.

§4. The Socle Obstruction

§4.1 Lemma 4.1 (Squarefree Span Lemma) and its socle reading. [PROVEN]

$$\mathbb{Z}\text{-span}\{1/p : p \text{ prime}\} = \{a/N \in \mathbb{Q} \text{ (lowest terms)} : N \text{ squarefree}\}.$$

Proof. ($\subseteq$) A finite sum $\sum a_p/p$ has common denominator $\prod p$, squarefree; divisors of squarefree integers are squarefree. ($\supseteq$) Integers lie in the span; for $N = p_1 \cdots p_r$ squarefree, Bézout gives $xq' + yp_1 = 1$ with $q' = p_2 \cdots p_r$, whence $1/N = x/p_1 + y/q'$, and induction closes. $\square$ (Checks G-13, G-14.)

Lemma 4.1' (socle reading). [PROVEN] In $\mathbb{Q}/\mathbb{Z}$, the subgroup generated by $\{1/p \bmod 1 : p \text{ prime}\}$ is exactly the socle — the set of elements of squarefree order — and the socle decomposes as $\oplus_p \mathbb{Z}/p$ [8].

An element $1/p^k \bmod 1$ has order p^k ; for $k \geq 2$ it lies outside the socle. (Check G-23.) The socle is precisely the multiplicative closure of the corpus transfer family computed in ZS-F27 Theorem 4.2, with Pontryagin dual $\prod_p \mathbb{F}_p$: the present lemma identifies that closure intrinsically, as the largest subgroup of $\mathbb{Q}/\mathbb{Z}$ reachable by prime-blind generation.

§4.2 Corollary 4.2 (Depth Exclusion). [PROVEN]

$1/p^k \notin \mathbb{Z}\text{-span}\{1/q : q \text{ prime}\}$ for every prime p and $k \geq 2$ (lowest-terms denominator p^k is not squarefree; check G-15).

§4.3 Theorem 4.3 (Prime-Blind Verticality, operator form). [PROVEN; v1.0 statement preserved]

Let K be any auxiliary space and V any invertible operator on K chosen independently of p . Any group of product-form operators $\{U(t) \otimes V^m\}$ on $\mathbb{C}^{11} \otimes K$ generated by elements $W_p \otimes V^{s_p}$ has register components $U(t)$ with $t \in \mathbb{Z}\text{-span}\{1/p\}$; hence none equals $U(1/p^k)$, $k \geq 2$. Proof: flow homomorphism (check G-11) + Lemma 4.1. $\square$

§4.4 Theorem 4.6 (Universal Prime-Blind No-Go; group-extension form). [PROVEN; new in v1.1]

Let G be any group and $\pi : G \rightarrow \mathbb{Q}/\mathbb{Z}$ any homomorphism. Suppose G is generated by a family $\{g_p : p \text{ prime}\} \cup A$ such that $\pi(g_p) = 1/p \bmod 1$ and every auxiliary generator $a \in A$ is prime-blind in the precise sense that $\pi(a)$ lies in the socle of $\mathbb{Q}/\mathbb{Z}$ (in particular, pure vertical generators with $\pi(a) = 0$ qualify). Then

$$\pi(G) \subseteq \text{socle}(\mathbb{Q}/\mathbb{Z}) = \oplus_p \mathbb{Z}/p,$$

and no element of G maps to an element of order p^k for any prime p and any $k \geq 2$.

Proof. The image of a homomorphism is the subgroup generated by the images of the generators; all generator images lie in the socle, which is a subgroup (Lemma 4.1'); elements of order p^2 are outside it. $\square$

Scope of the generalization. Theorem 4.6 removes every trace of the operator model: G may be non-abelian, the extension arbitrary (semidirect, central, wild), the auxiliary structure of any kind — only the existence of a register-time homomorphism and the prime-blindness of the auxiliaries are used. Theorem 4.3 is the instance $G \subseteq U(\mathbb{C}^{11} \otimes K)$, π = the register-time projection. Honesty note (hardest-step test, §9.1): the proof content is the socle structure of $\mathbb{Q}/\mathbb{Z}$ — textbook; the theorem's value is the clean delimitation of an extension class, not depth.

§4.5 Corollary 4.4' (Spectral Rigidity of Depth). [PROVEN · CORRECTED; defect of record in §8.3]

No conjugation on $\mathbb{C}^{11}$ carries any $U(t)$ with $t \in \mathbb{Z}\text{-span}\{1/p\}$ to $U(1/p^2)$.

Corrected proof. Conjugation preserves the eigenvalue multiset $\{e^{2\pi i t(j-5)} : j = 0, \dots, 10\}$, hence preserves the subgroup of $\mathbb{Q}/\mathbb{Z}$ generated by the exponents $\{t(j-5) \bmod 1\}$ and in particular its exponent, the maximal order. For $t = a/N$ in lowest terms, the $j = 6$ exponent t has order N , and every exponent has order dividing N , so the conjugation-invariant maximal order equals N . For $t \in \text{span}$, N is squarefree (Lemma 4.1); for $t = 1/p^2$, the maximal order is p^2 , not squarefree. $\square$ (Check G-22.)

Remark 4.4a (correction of record). The v1.0 proof asserted that the eigenvalue multiset determines $t \bmod 1$ up to the sign reflection $t \mapsto -t$. This step is FALSE: the multiset is invariant under the full unit action $t \mapsto ut$, $\gcd(u, N) = 1$ — explicit counterexample $\text{spec } U(1/11) = \text{spec } U(2/11)$, both equal to the complete set of 11th roots of unity (check G-21, constructed by adversarial audit). The conclusion of the corollary is unaffected; the corrected invariant is the maximal order, not the parameter. Per the no-deletion convention the defective step is recorded here and the protocol consequence in §8.3. The v1.0 numeric check G-16 verified the conclusion and is retained; it could not have caught the defect, because a numeric face verifies statements, not proofs — the gap §8.3 now closes.

§4.6 Corollary 4.5 (Universal-Multiplier Mismatch). [PROVEN; numeric face COMPUTED; v1.0 preserved]

The height engine of §3 supplies one rate, $-\log|\lambda| = 0.114835\dots$; the towers require the infinite family $\{\log p\}$. A single constant equals none of them (min gap 0.5783 at $p = 2$, check G-18; $|\lambda|$ vs $1/p$ min gap 0.3915, check G-17) and could equal at most one. The strong Vertical Conversion Hypothesis remains RETIRED in its entire prime-blind class, now under the wider Theorem 4.6.

§5. The Vertical Engine Identification (v1.0 preserved; tie to Theorem 4.6 added)

Theorem 5.1 (The Corpus Vertical is Archimedean). [DERIVED-interpretation] The Z-Spin iteration possesses exactly one canonical vertical axis (Theorem 3.3) with one universal rate; by Theorem 4.6 no prime-blind completion of the transfer family escapes the socle, so this verticality cannot supply finite-place depth. The single-axis, single-rate signature is that of the archimedean place, which retroactively explains the PROVEN content of ZS-F24/F25 (archimedean-complete, finite-place-open detection). Depth at the finite place p requires per-prime structure; the minimal simultaneous carrier of all primes is the restricted product — the adelic shape of the OPEN locator. Corollary 5.2 (O-F27.2 sharpened, DERIVED): depth one is forced, by the arithmetic of $\mathbb{Q}/\mathbb{Z}$ rather than by an axiom, for the entire prime-blind extension class; the remaining open content is prime-aware only. §5.3 (consistency with Finite Trace Rigidity) is preserved: the two no-go results close complementary routes by different mechanisms. New in v1.1: §6 identifies, by import, exactly what the per-prime structure must be — and shows it is smaller than v1.0 assumed.

§6. The Local Height Model (new in v1.1)

§6.1 Model 6.1 (Local Prime-Height Realization). [IMPORTED-known: primon gas / Bost–Connes local factor]

For each prime p let $H_p = \ell^2(\mathbb{N}_{\geq 1})$ with number operator $N|k\rangle = k|k\rangle$, and define the local height Hamiltonian

$$H_p = (\log p) \cdot N, \quad \text{Spec}(H_p) = \{k \log p : k \geq 1\}.$$

This is the single-prime factor of the primon gas of Julia and Spector [9, 10] and of the Bost–Connes Hamiltonian [11], whose bosonic partition function over all primes is $\zeta(s)$. It is recorded here at IMPORTED-known status: nothing in this section is claimed as new mathematics (NC-F29.8); what is new is only the registration of what the import does and does not close for the corpus programme.

§6.2 Proposition 6.2 (Support and Weight Realization). [DERIVED from IMPORTED-known]

The graded trace of the local factor with von Mangoldt insertion is the Euler logarithmic factor,

$$\text{Tr}_{\{H_p\}}((\log p) e^{\{-sH_p\}}) = \sum_{\{k \geq 1\}} (\log p) p^{-ks} = \log p \cdot p^{-s} / (1 - p^{-s}),$$

and the sum over primes assembles to the logarithmic derivative of zeta,

$$\sum_p \sum_{\{k \geq 1\}} (\log p) p^{-ks} = -\zeta'(s)/\zeta(s) \quad (\text{Re } s > 1).$$

Numeric faces: the closed form at machine precision for $p = 2, 3, 5, 7$ (check G-25); the prime-sum identity at $s = 2$ within 5×10^{-5} and at $s = 3$ within 10^{-8} against $-\zeta'/\zeta$ computed independently (check G-24, two evaluations; primes to 2×10^6). The model therefore realizes, simultaneously and exactly, the support $\{k \log p\}$ demanded by the locator target (ZS-F25 Thms 6.4-N/6.4-W; ZS-F27 Thm 5.1) and the von Mangoldt weight layer registered OPEN in v1.0 §8.

§6.3 Execution of the v1.0 Pre-Registered Factorization Audit. [PASS]

v1.0 §8 pre-registered, as the first audit of any future construction, the question whether the explicit-formula weight factorizes as (p-datum) $\times$ (level-datum). In Model 6.1 the answer is affirmative and exact: the insertion $(\log p)$ is constant in k , and the level dependence enters only through the spectrum — $\text{weight}(p, k) = \text{rate}(p) \times 1(k)$ (check G-26). Combined with the universality of the level space $\ell^2(\mathbb{N}_{\geq 1})$ (the same for every p), the audit sharpens Theorem 5.1's identification to its minimal form: the per-prime geometry the locator requires is not a per-prime space and not a per-prime shift — it is a per-prime rate, $\log p$, on one universal vertical.

§6.4 What the Import Closes and What It Relocates.

Model 6.1 closes the v1.0 open problems O-F29.1 and O-F29.2 at the model level: a prime-aware system with the correct support and the correct weights exists, is one line long, and is standard. It does not close them at the derivation level, and the relocation is the substantive content of this section: (i) the rates $\{\log p\}$ are inserted by hand — their corpus-internal derivation is the sharpened open problem O-F29.1'; (ii)

the model lives beside the register, not on it — no element acts as $U(1/p^2)$ on $\mathbb{C}^u$, consistent with Theorem 4.6 since the coupling H_p is prime-aware precisely through its rate; (iii) the contrast with §3 is now exact: the corpus archimedean vertical carries the one rate $-\log|\lambda| = 0.1148$, the finite-place verticals require the family $\{\log p\}$, and Corollary 4.5 is the statement that the former is not a member of the latter. The locator question, in its sharpest corpus form, is therefore: what corpus-internal principle, if any, produces the rate spectrum $\{\log p\}$? This is registered as O-F29.1' with the explicit non-claim NC-F29.9 that the present import constitutes no progress on it.

§6.5 Theorem 6.5 (Rate Spectrum No-Go). [PROVEN; new in v1.2 — the decidable core of the proposed F30 Stage 1]

(a) Torsion obstruction. Every element of the socle $\oplus_p \mathbb{Z}/p$ — the carrier of the corpus prime data (Lemma 4.1'; ZS-F27 Thm 4.2) — has finite (squarefree) order, while the additive reals are torsion-free. Hence $\text{Hom}(\text{socle}(\mathbb{Q}/\mathbb{Z}), (\mathbb{R}, +)) = 0$: for x of order N and any additive f , $N \cdot f(x) = f(Nx) = 0$ forces $f(x) = 0$. Every additive extraction of real rates from the corpus's finite-place order data vanishes identically.

(b) Finite-generation obstruction. $\sum a_p \log p = 0$ with integers a_p holds iff $\prod p^{a_p} = 1$ iff all $a_p = 0$ (unique factorization); hence $\{\log p : p \text{ prime}\}$ is $\mathbb{Q}$ -linearly independent and $\dim_{\mathbb{Q}} \text{span}_{\mathbb{Q}} \{\log p\} = \aleph_0$. (Numeric face: $\min |\sum a_p \log p| = 8.59 \times 10^{-5} > 10^{-6}$ over all nonzero integer vectors $|a_p| \leq 6$ on the first five primes, attained at $a = (-1, 2, -4, 5, -2)$; check G-29. Freeness with the Lindemann input is already corpus-registered as ZS-F27 Lemma 5.4.) Consequently no finite family of real constants — in particular no finite set of corpus-derived constants such as $-\log|\lambda|$ — can $\mathbb{Q}$ -linearly generate the rate spectrum. $\square$

Corollary 6.6 (negative closure of the declared class). [DERIVED] O-F29.1' is closed in the NEGATIVE for the class of homomorphic and finite- $\mathbb{Q}$ -linear extractions: relative to that class, the rate spectrum $\{\log p\}$ is necessarily imported as infinitely many independent data. The surviving residue, registered as O-F29.1'', is a nonlinear, spectral, or analytic derivation from infinite corpus-internal structure — which, by Theorem 5.1, is the locator construction itself. This is precisely the Stage-1 verdict the proposed F30 line requested (“prove that corpus-internal finite/socle structures cannot generate logarithmic rates”), obtained inside the existing line per the drafting gate, without opening a new one.

§6.6 The Semilocal Height Ansatz and Its Three Gates. [registration: DERIVED restatement; no gate closed]

Define, for a finite set of primes S , the semilocal height operator $H_S = H_{\infty} \oplus \oplus_{p \in S} (\log p) \cdot N$ on $H_S = H_{\infty} \oplus \oplus_{p \in S} \ell^2(\mathbb{N}_{\geq 1})$, where H_{∞} is a realization of the archimedean place — at the kinematic level the Koenigs–Abel vertical of §3, at the analytic level the scaling action of [4, 6]. By Proposition 6.2 restricted to S , the finite-place block reproduces the explicit-formula support $\{k \log p : p \in S, k \geq 1\}$ and the von Mangoldt weights exactly, and schematically, in the Weil normalization of [4, 6],

$$W_S(g) = W_{\infty}(g) + \sum_{p \in S} \sum_{k \geq 1} (\log p) p^{-k/2} [g(k \log p) + g(-k \log p)] - \text{pole terms}.$$

The RH-relevant content of the ansatz reduces to three gates: SG1 — finite- S positivity, $W_S(g) \geq 0$ for every finite S ; SG2 — positivity-preserving convergence $W_S \rightarrow W_{\zeta}$; SG3 — density of the test class

for the Weil criterion. Mapping onto the registered walls: SG1 is the semilocal frontier of [6] and, in corpus coordinates, the window-ladder rung structure of ZS-F26 (rung 1 $\equiv$ MI(2) $\equiv$ sup_N $\varepsilon^*(N) = 0$); SG2 is the finite $\rightarrow$ infinite Euler passage registered at O-F26.5 and now also as O-F29.5; SG3 is classical. Status: the ansatz is a coordinate system — Model 6.1 $\oplus$ archimedean — not a construction beyond it; no gate is closed, advanced, or claimed here (NC-F29.10). This paragraph is the registration the first audit's strongest single-paper proposal requested, at the only grade the hardest-step test permits.

§6.7 Adjudication of the Proposed F30–F33 Lines. [audit of record; one refutation PROVEN]

Four RH-grade lines were proposed by the first further audit. They are adjudicated here under the drafting gate (a paper only when reconnaissance yields theorems) and the second audit's calibration (direct RH attack: probability ≈ 0 , with an external-credibility cost that would damage the live track).

F30 (uniform finite-to-infinite positivity-preserving convergence $Q_P \rightarrow Q_\zeta$). Declined as a paper: its target is gate SG2 verbatim — the recognized hard core of the external programme — and no corpus tool addresses it. Its Stage-1 sub-question (rate derivation or no-go) was decidable now and is decided above as Theorem 6.5. F31 (semilocal prolate attachment). Declined as a paper: the attachment point exists and is registered (§6.6, $H_\infty \oplus$ finite-place blocks); the positivity content is layer E, untouched by this line. F32 (determinant convergence with a height-regularized continuation lemma). Declined as a paper: the logarithmic-derivative layer it needs is Proposition 6.2; the proposed continuation across the critical strip is beyond current tools and is registered as the analytic face of SG2, OPEN.

Proposition 6.7a (refutation of the naive total-positivity route, F33). [PROVEN] The proposed target — $\sum_{\{i,j\}} c_i c_j K_S(t_i - t_j) \geq 0$ for all finite S , with K_S the symmetrized finite-place kernel $\sum_{\{p \in S, k \geq 1\}} (\log p) p^{-k/2} [\delta(t - k \log p) + \delta(t + k \log p)]$ — is false. Two-point counterexample: K_S has zero diagonal mass (no atom at 0), so with $S \ni 2$, $t = (0, \log 2)$, $c = (1, -1)$, the form equals $2c_1 c_2 \cdot (\log 2) 2^{-1/2} = -\sqrt{2} \log 2 = -0.9803 < 0$ (check G-27). Smoothing does not rescue the route: the Bochner transform of the smoothed kernel $K_S * h_\varepsilon$ is $\hat{h}_\varepsilon(\xi) \cdot f_S(\xi)$ with $\hat{h}_\varepsilon \geq 0$ and $f_S(\xi) = \sum 2(\log p) p^{-k/2} \cos(k \xi \log p)$ sign-indefinite — grid minima -55.6 at $\xi = 0.60$ for $S = \{p \leq 500\}$ and -125.5 at $\xi = 0.50$ for $S = \{p \leq 2000\}$ (check G-28) — so the smoothed kernel is not positive-definite either (Bochner–Schwartz). Structural reductio: were the prime side alone positive-definite, Weil positivity would reduce to the archimedean-plus-pole terms and would not be the open frontier it is; the sign-indefiniteness of the prime side is the entire content of the problem. The total-positivity route survives only as the full-kernel statement — which is layer E again. $\square$

§7. The Six-Layer Ladder, Gated (upgraded from the v1.0 consolidation table)

v1.0 §6 resolved the “same versus different” debate by decomposition. v1.1 upgrades the table into an evaluation framework: each layer now carries a gate — the statement a candidate locator model must satisfy at that layer — so that the ladder functions as a hierarchical no-go / yes-go test battery for RH-locator proposals, internal or external. [DERIVED-interpretation: the gates restate registered results; no entry adds a claim.]

Table 7.1. The gated Six-Layer Ladder. “Gate” states what a candidate model must exhibit at that layer; “Status” is the corpus-registered state after this release.

Layer	Gate (what a model must exhibit)	Status after v1.1	Source
A. Frame	A $\mathbb{Z}_2$ involution whose fixed locus carries the critical parameter ($\sigma = 1/2 \leftrightarrow j = 1/2$).	SHARED (DERIVED-interpretation)	ZS-M28 Thm 28.5
B. Detection	A scaling flow detecting the archimedean place; here: the Koenigs–Abel height with unit shift (pure complex dynamics).	SHARED; PROVEN kinematic core	§3; ZS-F24/F25
C. Support / depth	Escape from the socle: some element of order p^k , $k \geq 2$, in the register-time image. Gate theorem: impossible for every prime-blind extension (Thm 4.6).	CLOSED against prime-blind class (PROVEN); realized at model level by import (§6); homomorphic rate extraction from socle data impossible (Thm 6.5a)	Thm 4.6; Model 6.1; Thm 6.5
D. Weights	von Mangoldt scale $\log p$ with $\zeta_p(k)$ profile; factorization rate(p) $\times$ level(k).	CLOSED at model level ($-\zeta'/\zeta$, audit PASS); derivation closed NEGATIVE for homomorphic / finite- $\mathbb{Q}$ -linear extraction (Thm 6.5); residue O-F29.1”	Prop. 6.2; §6.3–6.5
E. Location / positivity	Weil positivity ladder; rung 1 $\equiv$ MI(2) $\equiv$ $\sup \epsilon^*(N) = 0$; internal V_4 -scalar route excluded.	OPEN; internal scalar route PROVEN-closed	ZS-F26 §7.7; ZS-M22 ADS-5/6
F. Count axis	An arithmetic count axis shared with the zero count; the dynamical height α exists but is prime-blind and of undecided observability.	Sharpened NON-CLAIM	§8.2; ZS-F13 Thm F13.2

Reading. “The same” remains true at layers A–B; “different” remains true at C (prime-blind, PROVEN), at D’s derivation level, and at E. The honest terminus of ZS-F24 has not moved. What v1.1 adds is operational: any future locator proposal — corpus-internal or external — can be scored layer by layer against Table 7.1, with C and D now carrying both a no-go theorem and a minimal yes-go model that delimit the admissible designs from below and above.

v1.2 update to the reading. Layer E now carries a registered external interface: the translation bridge between the corpus rung quantity $\sup_N \epsilon^*(N)$ (ZS-F26 §7.7) and the small-eigenvalue / eigenvector objects of the external truncated-Weil-form and prolate programme [6, 7, 12] — the same object in internal and external coordinates. Under NC-F29.10’s discipline the bridge is a contribution of coordinates, not progress; its execution belongs to the verification track of §13.

§8. Corrections and Sharpenings of Record

§8.1 The ZS-M23 §12 Principle, Refined. [v1.0 preserved]

“Z-Spin does not, and structurally cannot, prove RH from internal data alone” stands; its PROVEN basis covers three audited route classes — finite-dimensional flows (ZS-F27 Thm 5.5), V_4 -equivariant scalars (ADS-5/6), and prime-blind vertical augmentations (now in the wider Theorem 4.6 form). Routes outside the classes (prime-aware, O-F29.1’) remain OPEN, and since layers D (derivation level) and E remain in any case, even their success would not by itself contradict the principle’s conclusion.

§8.2 NC-M28.7, Sharpened. [v1.0 preserved; erratum candidate for ZS-M28 v1.x]

A dynamical count axis exists (α , Theorem 3.3) but is prime-blind (Theorem 4.6) and not shown corpus-observable (the three failed candidates of F13.2 remain failed). NC-M28.7 survives as: “no shared arithmetic count axis; a dynamical height axis exists but is prime-blind and of undecided corpus observability.” Corollary 28.5a is scope-limited to the horizontal component. No deletion.

§8.3 Corollary 4.4: Proof-Step Correction of Record and the Protocol Consequence. [new in v1.1]

Defect. The v1.0 proof of Corollary 4.4 used the step “the eigenvalue multiset of $U(t)$ determines $t \bmod 1$ up to sign,” which is false: the multiset is invariant under the unit action $t \mapsto ut$, and at $N = 11$ the orbit is everything — $\text{spec } U(1/11) = \text{spec } U(2/11)$ (check G-21). The step was found by adversarial counterexample construction during the second independent audit; the conclusion is true and the corrected proof (§4.5, via the conjugation-invariant maximal order) is simpler than the defective one.

Why the gap existed. The v1.0 verification suite checked the corollary’s conclusion (G-16) and passed, because the conclusion is true. A numeric face verifies statements, not proofs: a true theorem with a false proof passes every conclusion-level check. The same gap admitted a defect in a parallel-track paper in the same season. Protocol consequence, adopted from this release onward (§9.2): every PROVEN tag must additionally pass a proof-step adversarial audit — a separate pass in which, for each step of the proof, a counterexample to the step itself (not to the theorem) is actively sought — performed before release; and the load-bearing lemmas of each paper are registered as machine-formalization candidates. For this paper: Lemma 4.1/4.1’ and Theorem 4.6 are registered as Lean/Coq candidates (elementary, fully formalizable); Corollary 4.4’s corrected proof passed the adversarial pass (the maximal order is conjugation-invariant by the spectral theorem; the unit action preserves it; no further rigidity is claimed).

§9. Anti-Numerology and Density Audit

§9.1 Anti-Numerology; Hardest-Step Test; Imported/New Ratio.

Zero free parameters: T , z^* , λ , φ , α are canonical (§3); the socle, the primes, and $\mathbb{Q}/\mathbb{Z}$ are canonical arithmetic (§4); Model 6.1 contains no parameter (the rates $\log p$ are the import’s defining data, not a fit). $A = 35/437$ is not consumed numerically; $Q = 11$ enters only as register dimension. The values $|\lambda|$ and $-\log|\lambda|$ are used solely to establish mismatch — the anti-numerological direction. Hardest-step self-report (density protocol): the most difficult single ingredient in this paper is, by honest assessment, the socle structure of $\mathbb{Q}/\mathbb{Z}$ or Koenigs 1884 — both textbook. By the audit’s own criterion this paper is therefore a map, a correction, and an import, and is labeled as such (§1). Imported/new weight ratio in the proof chains: §3 $\approx 90\%$ imported (Koenigs) / 10% new (application + numerics); §4 $\approx 60/40$ (socle imported; the extension-class delimitation, the operator instance, and the corrected spectral corollary new in formulation); §6 $\approx 95/5$ (primon gas imported; only the registration logic new). No section’s value claim exceeds its new fraction.

§9.2 Proof-Step Adversarial Audit (executed for this release).

Every PROVEN tag in this paper passed a step-level adversarial pass: Lemma 4.1 (both inclusions probed: the squarefree-reduction step against composite-denominator targets, G-15; the Bézout step constructively, G-14); Lemma 4.1' (order computations, G-23); Theorem 4.3/4.6 (the homomorphism step, G-11; the generated-subgroup step is definitional); Corollary 4.4' (the maximal-order invariance step against the unit action that killed v1.0's step, G-21/G-22); Corollary 4.5 (the single-constant step is logical; gaps numeric, G-17/G-18). Formalization candidates registered: Lemma 4.1/4.1', Theorem 4.6.

§10. Multilayer Falsification Gates

Table 10.1. Falsification gates for ZS-F29 v1.2. Layers: mathematical breakdown (F-F29.1, .2, .3, .5, .9), external dependency / external events (F-F29.5, .9, .13), structural integrity (F-F29.4, .7, .11, .14), anti-overclaim (F-F29.6, .8, .10, .12). All gates currently PASS.

Gate	Falsification condition (triggers if TRUE)	Consequence
F-F29.1	Re-execution of G-06–G-08 fails (Koenigs / Abel violated beyond tolerance).	Thms 3.1/3.3 retracted; script re-audited (would contradict Koenigs 1884).
F-F29.2	An explicit prime-blind construction (now: any G , π per Thm 4.6) exhibits register-time image of order p^2 .	Contradicts Lemma 4.1'; immediate re-audit of prime-blindness. Registered for completeness.
F-F29.3	$ \lambda = 1/p$ or $-\log \lambda = \log p$ at all precision.	Cor. 4.5 collapses. (Excluded at 120 digits, G-17/G-18.)
F-F29.4	A corpus observable for α is constructed without per-prime structure and carries arithmetic content.	§8.2 sharpening wrong; NC-M28.7 and F13.2 jointly re-audited.
F-F29.5	Koenigs inapplicable at z^* ($\lambda = 0$ or $ \lambda \geq 1$ at higher precision).	Thm 3.1 collapses. (Excluded, G-03/G-20.)
F-F29.6	A prime-aware construction achieves towers but the $\text{rate}(p) \times \text{level}(k)$ factorization is proven impossible for it.	The §6.3 audit standard fails for that route; layer-D gate revised.
F-F29.7	Any entry of Table 7.1 contradicts the registered status in its source paper.	Consolidation error; table re-audited.
F-F29.8	A hidden free parameter is found in §3–§6.	Zero-free-parameter claim invalidated; anti-numerology flag.
F-F29.9	The Model 6.1 trace identity fails numerically on re-execution (G-24/G-25), or the import [9–11] is shown misattributed.	§6 re-audited; import citation corrected.
F-F29.10	Model 6.1 is presented anywhere downstream as a corpus derivation of $\{k \log p\}$ or of $\log p$ weights.	Anti-overclaim breach of NC-F29.8/NC-F29.9; demote and flag.
F-F29.11	A further false proof step is found under adversarial audit in F24–F29.	Systematic step-level re-audit of every PROVEN tag in the arc; results published as corrections of record.
F-F29.12	Any downstream corpus document frames F29-line content, or a successor line, as an RH attempt or as RH-proof-grade without first passing the hardest-step test and the probability $\times$ value calibration of §13.	Anti-confabulation breach; the document is demoted and flagged; programme note re-audited.

F-F29.13	SG1 (finite-S Weil positivity including finite places) is established externally for all finite S.	§6.6 mapping re-audited; layer-E gate updated; the ansatz registration upgraded to a live interface.
F-F29.14	A corpus-internal nonlinear or spectral derivation of $\{\log p\}$ is exhibited.	Theorem 6.5's class boundary confirmed or re-audited; O-F29.1" closes positive; layer-D gate updated.

§11. Open Problems

O-F29.1' $\rightarrow$ O-F29.1" (the rate spectrum; status updated in v1.2). The declared-class branch of O-F29.1' is now CLOSED in the negative: by Theorem 6.5, no homomorphic extraction from the socle data and no finite- $\mathbb{Q}$ -linear combination of corpus constants produces $\{\log p\}$ — within that class the rates are necessarily imported as infinitely many independent data. The surviving open problem O-F29.1": a nonlinear, spectral, or analytic derivation of the rate family from infinite corpus-internal structure — which, by Theorem 5.1, is the locator construction itself. Sub-target preserved: realization of $U(1/p^2)$ on the register by a declared prime-aware (rate-carrying) generator.

O-F29.2' (the $\zeta_p(k)$ profile). Model 6.1 realizes the geometric weights $p^{\{-ks\}}$ with constant insertion $\log p$; the corpus normalization of ZS-F25 Thms 6.4-N/6.4-W specifies a $\zeta_p(k)$ profile whose match to the model's grading must be established or refuted at the derivation level.

O-F29.3 (observability of the height; unchanged). Decide whether α is a corpus observable, against the three failed candidates of F13.2.

O-F29.4 (the height-graded trace; unchanged). Compute height-graded traces of the coupled archimedean system and decide whether any new invariant below depth appears.

O-F29.5 (the SG2 gate; registered, not attacked; new in v1.2). Positivity-preserving convergence $W_S \rightarrow W_\zeta$ of the semilocal Weil forms (§6.6) — the finite $\rightarrow$ infinite Euler passage of the external programme, registered in corpus coordinates at O-F26.5; OPEN; explicitly outside the tools of this line (§6.7).

§12. Non-Claims

NC-F29.1. No claim on RH or GRH; Gate D5 remains IMPORTED-OPEN $\equiv$ RH.

NC-F29.2. No claim that the prime-aware derivation (O-F29.1') is achievable; Theorem 5.1 and §6.4 identify its shape, not its existence.

NC-F29.3. No claim that the height α is a corpus observable (O-F29.3).

NC-F29.4. The gated ladder (§7) is consolidation plus restatement: no entry adds a claim beyond its source.

NC-F29.5. No new free parameter; $A = 35/437$ and $Q = 11$ LOCKED and unchanged; A not consumed numerically.

NC-F29.6. The refutation of the strong Vertical Conversion Hypothesis concerns the prime-blind class only (now in the Theorem 4.6 generality); it is not evidence against prime-aware routes.

NC-F29.7. The conversational and founding-note motivations (§1, Remark 3.4 of v1.0, Appendix B) carry no evidentiary weight and are used in no proof.

NC-F29.8. Model 6.1 and Proposition 6.2 are known mathematics (primon gas / Bost–Connes [9–11]); no novelty is claimed for any statement of §6; the external retitling of this release positions the paper and does not upgrade its genre.

NC-F29.9. The model-level closure of O-F29.1/O-F29.2 is not a corpus derivation and constitutes no progress on O-F29.1"; the rates $\{\log p\}$ are inserted, not derived.

NC-F29.10. The semilocal height ansatz registration (§6.6) and the gated mapping are coordinates, not progress: no gate SG1–SG3 is closed, advanced, or claimed; the layer-E translation bridge is likewise a contribution of coordinates only.

NC-F29.11. The adjudication of §6.7 declines F30–F32 as papers on programme grounds; except where proven (the declared class of Theorem 6.5; Proposition 6.7a), no mathematical impossibility is claimed for those lines.

NC-F29.12. Nothing in the F29 line is, or may be cited as, an attempt on RH; gate F-F29.12 makes this binding on successors.

§13. Conclusion and Programme Note

Two audits, one season apart from the reconnaissance that produced v1.0, have shaped this closing release. The first asked the paper to face outward; it now does, at the only honest price — by importing what is known (the primon-gas local factor) rather than rediscovering it, and by widening its one real obstruction to the cleanest class it governs (every prime-blind extension, Theorem 4.6). The second audit found a false step inside a true theorem and, more importantly, named the gap that let it through: conclusion-level verification cannot see proof-level defects. That gap is now closed by protocol (§8.3, §9.2), and the defect, its counterexample, and its correction are part of the record.

The mathematical residue of the F29 line is small and exact: the Z-Spin iteration is a unit shift on the archimedean vertical; no prime-blind structure escapes the socle; the locator's missing engine is precisely the rate spectrum $\{\log p\}$ on one universal level space — not a space, not a shift, a family of rates — and the corpus owns exactly one rate. The Six-Layer Ladder, now gated, turns this residue into an evaluation instrument for any future locator proposal. Programme note (registered): the F29 line completes here; by the focus audit, the collaboration's external effort is redirected to the seam-constrained Fuglede track, where the corpus's only currently live external theorem programme resides. The honest terminus has not moved; its front door now has a shape, a gate battery, and a posted price.

Programme portfolio (v1.2, registered). Per the second further audit’s calibration — direct RH attack: probability ≈ 0 with an external-credibility cost; independent verification of the live frontier: the historically attested entrance — the collaboration’s allocation is set at approximately 80% to the seam-constrained Fuglede track (the only currently live external-theorem programme) and approximately 20% to an independent-verification interface with the truncated-Weil-form / prolate programme [6, 7, 12]: exact-arithmetic reproduction, basis benchmarking, convergence-rate measurement of zero approximants, and precise discrepancy reporting, in Z-Spin-free language — precisely the competency this line demonstrated (120-digit verification, exact rational arithmetic, adversarial proof-step audit). The translation bridge of §7 ($\sup \epsilon^*(N) \leftrightarrow$ small eigenvalues) is the single registered corpus $\leftrightarrow$ frontier interface, under NC-F29.10’s discipline. The four proposed RH-grade lines are adjudicated in §6.7; gate F-F29.12 makes the calibration binding on successors. Breakthroughs, where they have come, have come from perfecting small tools, not from aiming at the big problem; this paper closes by pointing its small tools at the frontier’s verification layer and its theorems at the walls they can actually reach.

Acknowledgements & Code Availability

This work was developed with the assistance of AI tools (Anthropic Claude, OpenAI ChatGPT, Google Gemini) for mathematical verification, code generation, and manuscript drafting, including two independent AI audit passes whose findings shaped v1.1; the adversarial counterexample of §8.3 was produced in audit. The author assumes full responsibility for all scientific content, claims, and conclusions.

Code availability. Verification script: `zs_f29_verify_v1_2.py` (mpmath 120-digit + exact rational arithmetic; deterministic under pre-registered seed 20260612; prime sieve to 2×10^6). Expected output: 30/30 PASS. All scripts will be publicly available at <https://github.com/KennyKang-git/zspin> upon release.

Appendix A. Verification Suite (30/30 PASS)

Table A.1. ZS-F29 v1.2 verification checks. G-01–G-20 reproduce the v1.0 suite; G-21–G-26 added in v1.1 (correction of record; socle face; Local Height Model); G-27–G-29 added in v1.2 (F33-naive refutation faces; $\mathbb{Q}$ -independence face).

Check	Statement	Result
G-01–05	z^* fixed point, multiplier consistency, $ \lambda \in (0,1)$, two-step closure, corpus value match (v1.0).	PASS
G-06–09	Koenigs nonvanishing; functional equation (3.6×10^{-47}); Abel increment (9.0×10^{-47}); height divergence at z^* .	PASS
G-10–12	Dirichlet-kernel trace; flow homomorphism; $\text{order}(W_p) = p$ for $p \leq 13$.	PASS
G-13–15	Squarefree span: 100,000 random reductions; Bézout realization; 200,000-sample depth-target exclusion.	PASS
G-16	$\text{spec } U(1/p^2) \neq \text{spec } U(a/N)$, N squarefree ≤ 30 , $ a \leq 30$ ($p = 2, 3, 5$) — conclusion-level check retained.	PASS
G-17–18	$ \lambda - 1/p \geq 0.3915$; $ - \log \lambda - \log p \geq 0.5783$.	PASS

G-19–20	$D = \text{diag}(j-5)$ sanity; vertical shift aperiodic.	PASS
G-21	Counterexample to the v1.0 Cor 4.4 proof step: $\text{spec } U(1/11) = \text{spec } U(2/11)$ exactly (unit action).	PASS (equality holds)
G-22	Corrected invariant: maximal exponent order = N for $U(a/N)$, N squarefree ≤ 30 ; $= p^2$ for $U(1/p^2)$, $p = 2, 3, 5$.	PASS
G-23	Socle face: order in $\mathbb{Q}/\mathbb{Z}$ of 5,000 sampled span elements is squarefree.	PASS
G-24	$\sum_p \log p/(p^s - 1) = -\zeta'(s)/\zeta(s)$: $s = 2$ within 5×10^{-5} , $s = 3$ within 10^{-8} (primes to 2×10^6 ; two checks).	$\text{PASS} \times 2$
G-25	Single-prime closed form, $p = 2, 3, 5, 7$, $K = 220$, tolerance 10^{-100} .	PASS
G-26	Pre-registered factorization audit: $\text{weight}(p, k) = \text{rate}(p) \times l(k)$, constant in k .	PASS
G-27	Two-point counterexample to finite-place kernel positive-definiteness: form value $-\sqrt{2} \log 2 = -0.9803 < 0$ (zero diagonal mass).	PASS
G-28	Truncated prime-side cosine transform sign-indefinite: grid minima -55.6 ($S = \{p \leq 500\}$, $\xi = 0.60$) and -125.5 ($S = \{p \leq 2000\}$, $\xi = 0.50$).	PASS
G-29	$\mathbb{Q}$ -independence face: $\min \sum a_p \log p $ over nonzero $ a_p \leq 6$ on the first five primes $= 8.59 \times 10^{-5} > 10^{-6}$, at $a = (-1, 2, -4, 5, -2)$.	PASS

Appendix B. The Point of Infinite Height (OBSERVATION; v1.0 preserved)

Since $\varphi(z^*) = 0$, the height diverges at the fixed point: $\alpha(z) \rightarrow \infty$ as $z \rightarrow z^*$ (check G-09). The unique attractor of the Z-Spin iteration is the point of infinite height — recorded as the dynamical address of the exploration-session picture that motivated the hypothesis, and as a structural echo of the corpus reading “a point that is also a size; a center that is also a boundary.” Status: OBSERVATION; used in no proof; no evidentiary weight (NC-F29.7).

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[ZS-M22] K. Kang, “Dedekind Zeta Chains and the V_4 Factorization,” ZS-M22 (2026), ADS-5/ADS-6.
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[ZS-M28] K. Kang, “The Z_2 -Involution Synthesis,” ZS-M28 v1.0 (2026), Thm 28.5, Cor. 28.5a, NC-M28.5–7.
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Version History

v1.0 (March 2026): Initial public release. Height Coordinate Theorem (Thms 3.1/3.3, DERIVED from Koenigs 1884; Abel unit shift; aperiodicity); Squarefree Span Lemma and Depth Exclusion (PROVEN); Prime-Blind Verticality Theorem with spectral-rigidity and universal-multiplier-mismatch corollaries (PROVEN) — the strong Vertical Conversion Hypothesis registered against O-F27.2 RETIRED in its prime-blind class; Vertical Engine Identification (Thm 5.1: the corpus vertical is archimedean; the missing engine $\equiv$ the adelic locator); Six-Layer Ladder consolidation; ZS-M23 §12 refinement and NC-M28.7 sharpening (erratum candidates); weight layer registered OPEN with pre-registered factorization audit; gates F-F29.1–8; open problems O-F29.1–4; Appendix B observation. Verification 20/20 PASS. Zero free parameters. No claim on RH. (Consolidated from internal Z-Spin Collaboration research notes up to the Vertical Conversion reconnaissance session, v0.1 $\rightarrow$ v1.0.)

v1.1 (March 2026): Closing release following two independent audits; all v1.0 statements preserved per the no-deletion convention. (a) Correction of record: the v1.0 Corollary 4.4 proof step “multiset determines t up to sign” found FALSE by adversarial counterexample $\text{spec } U(1/11) = \text{spec } U(2/11)$ (check G-21); conclusion unaffected; corrected proof via the conjugation-invariant maximal order (Cor. 4.4', check G-22); proof-step adversarial audit added to the verification standard with Lemma 4.1/Thm 4.6 registered as Lean/Coq formalization candidates (§8.3, §9.2). (b) Generalization: the obstruction promoted from the operator instance to the Universal Prime-Blind No-Go on arbitrary groups via the socle of $\mathbb{Q}/\mathbb{Z}$ (Lemma 4.1', Theorem 4.6, PROVEN; check G-23), with the explicit tie to ZS-F27 Thm 4.2 ($\oplus_p \mathbb{Z}/p$, dual $\prod_p \mathbb{F}_p$). (c) Constructive counterpart by honest import: the Local Height Model (Model 6.1, IMPORTED-known: primon gas / Bost–Connes [9–11]) realizing support $\{k \log p\}$ and von Mangoldt weights simultaneously, with $\Sigma = -\zeta'/\zeta$ (Prop. 6.2; checks G-24/G-25); the v1.0 pre-registered factorization audit executed with result PASS (weight = $\text{rate}(p) \times \text{level}(k)$, check G-26); O-F29.1/O-F29.2 closed at model level and relocated to the rate-derivation problem O-F29.1' and the profile problem O-F29.2', with non-claims NC-F29.8/NC-F29.9 barring overclaim. (d) The Six-Layer Ladder upgraded to a gated evaluation framework (Table 7.1). (e) External-first retitle with Z-Spin confined to the subtitle (positioning only; genre unchanged per §9.1). (f) Density audit added: hardest-step self-report (textbook), imported/new ratios per section. (g) Gates F-F29.9–11 added; programme note registered — the F29 line

completes at v1.1; external effort redirected to the seam-constrained Fuglede track. Verification 27/27 PASS (zs_f29_verify_v1_1.py). Zero free parameters; $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED unchanged; A not consumed numerically. No claim on RH; Gate D5 remains IMPORTED-OPEN $\equiv$ RH. (Consolidated from internal Z-Spin Collaboration research notes; v1.1 executing the two-audit prescription.)

v1.2 (March 2026): Terminal addendum to the closing release, executing the decidable core of two further audits; all v1.0/v1.1 statements preserved per the no-deletion convention. (a) Rate Spectrum No-Go (Theorem 6.5, PROVEN): $\text{Hom}(\text{socle}(\mathbb{Q}/\mathbb{Z}), (\mathbb{R}, +)) = 0$ (torsion vs torsion-free) and $\dim_{\mathbb{Q}} \text{span}_{\mathbb{Q}} \{\log p\} = \aleph_0$ (unique factorization; numeric face G-29: min gap 8.59×10^{-5} ; freeness previously registered as ZS-F27 Lemma 5.4) — within the class of homomorphic and finite- $\mathbb{Q}$ -linear extractions the rate family is necessarily imported; O-F29.1' closed negative in that class, residue O-F29.1" registered; the Stage-1 question of the proposed F30 line thereby decided inside the existing line per the drafting gate. (b) Semilocal Height Ansatz registered with gates SG1–SG3 mapped onto ZS-F26 rung 1 / O-F26.5 ($\equiv$ new O-F29.5) / classical density (§6.6; registration, not progress; NC-F29.10). (c) Adjudication of the proposed F30–F33 lines (§6.7): F30–F32 declined as papers and absorbed as gates; the naive total-positivity route (F33) REFUTED (Proposition 6.7a, PROVEN: two-point counterexample $-\sqrt{2} \log 2$, check G-27; Bochner sign-indefiniteness of the smoothed kernel, check G-28; structural reductio). (d) Six-Layer Ladder layers C–D updated; layer-E external interface registered ($\sup \varepsilon^*(N) \leftrightarrow$ small-eigenvalue translation bridge, [6, 7, 12]). (e) Programme portfolio registered ($\approx 80\%$ seam-constrained Fuglede track / $\approx 20\%$ independent-verification interface); gates F-F29.12–14 added, with F-F29.12 the anti-confabulation gate binding on successors. Verification 30/30 PASS (zs_f29_verify_v1_2.py). Zero free parameters; $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED unchanged; A not consumed numerically. No claim on RH; Gate D5 remains IMPORTED-OPEN $\equiv$ RH. (Consolidated from internal Z-Spin Collaboration research notes; v1.2 executing the F30–F33 adjudication.)